

Introduction to Database Systems (Spring 2025)

Section	Lecture Time and Room	Department
CSC 675-775-01	Thur 7:00-9:45PM BH 352	Computer Science

Instructor: Joseph Hui, jhui@sfsu.edu

Course Website: All of the course materials will be available through Canvas. <http://canvas.sfsu.edu>

Office Hours: By appointment only (email and I'll schedule a time. Zoom link on Canvas).

CSC 675-775 is a paired course for undergraduate and graduate students. Undergrad students must enroll in CSC 675 and graduate students must enroll in CSC 775.

Prerequisites:

- CSC 675: CSC 413 with a grade of C or better.
- CSC 775: graduate standing or consent of the instructor.
- Conditionally classified grads who had CSC413 as a condition must clear this first before enrolling in CSC775
- Students who have completed CSC 675 may not take CSC 775 for credit.
- The prerequisites for this course are strictly enforced for all sections. Concurrent enrollment is not allowed.

Recommended Reading: In this class you don't need a textbook. But, if you need a good reference, then I recommend the following book: **Avi Silberschatz, *Database System Concepts, 7th or later***

If you are also interested in the implementation of RDBMS, then I recommend the following book: **Michael Stonebraker, *Readings in Database Systems, 3rd edition***

Objectives: Design and architecture high quality and consistent relational database systems from concept/idea to integration, testing and delivery. Describe, define and apply the major components of the relational database model to database design. Learn and implement the principles and concepts of information integrity, security and confidentiality. Apply ethical computing concepts and practices to database design and implementation.

Tentative Course Outline:

- Semantic/Conceptual model: Use cases, Database Requirements, Entity Relationship Diagrams (ERD)
- Relational model: Database model/EER, forwarding engineering, SQL, triggers, stored procedures, functions, normalization
- Data Storage, Security, File Organization, and Indexing
- Query Optimization, Evaluation and External Sorting
- Transaction Management and Concurrency Control
- Accessing Databases with Java and ORM Frameworks

Grading Policy:

1. Four Milestones 50%
 - a. Milestone 1 (10%) Product Summary with Database Requirements
 - b. Milestone 2 (10%) Database Conceptual Design (ERD)
 - c. Milestone 3 (10%) Database Architecture (EERD and SQL)
 - d. Milestone 4 (10%) Programmatic access (Java program via JDBC)
 - e. Milestone 5 (10%) Object/Relational Mapping in Java
2. Midterm Exam I: 25%
3. Midterm Exam II: 25%
4. Final Exam (optional): 25% – best two (2) scores from Midterm 1, Midterm 2, and Final

Tentative Schedule:

Week	Date	Topic	Assignments
1	1/30	Syllabus, Course Policies, and Intro to Relational Databases	Getting Started Module Milestone #1 Assigned
2	2/06	Use Cases and Database Requirements	
3	2/13	The Conceptual Model Entity Relationship Diagram (ERD)	
4	2/20	ERDs II Introduction to SQL	Milestone #1 Due, Milestone #2 Assigned
5	2/27	Relational Database Management Systems	
6	3/06	Indexing Review for Midterm I	Milestone #2 Due
7	3/13	Midterm I exam	
8	3/20	Intermediate SQL Advanced SQL	Milestone #3 Assigned
9	3/27	No Class (Spring Break)	
10	4/3	Transaction Management Concurrency Control	Milestone #3 Due, Milestone #4 Assigned
11	4/10	Accessing Databases with Java I	
12	4/17	Accessing Databases with Java II Object Modeling Frameworks	
13	4/24	Database Internals	Milestone #4 Due Milestone #5 Assigned
14	5/1	Midterm II exam	
15	5/8	Performance & Tuning	
16	5/15	Guest Lecturer Final Review	
17	5/22	Final Exam	Milestone #5 Due

Communication:

- Email is the best way to communicate with me outside of class.
- Feel free to send me questions about the software industry or your resume for feedback.
- This is a live document; I may make changes to any of these policies and contents at any time during the semester. I will announce changes in class and post them on Canvas.

Important Dates:

1. Midterm I Exam:
 - a. Thursday, Mar 13, 7-9:45pm
2. Midterm II Exam
 - a. Thursday, May 1, 7-9:45pm
3. Final Exam:
 - a. Thursday, May 22, 7:15-9:15pm

Deadlines: All the assignments will be due on Thursdays at 7:00pm. Please read the course policies to learn more about the policies for checkpoint assignments in this course.

Grades: The final letter grade for the course will be computed using the following distribution:

Highest	Lowest	Letter Grade
100	93	A
92.99	90	A-
89.99	87	B+
86.99	83	B
82.99	80	B-
79.99	77	C+
76.00	73	C
72.99	70	C-
69.99	67	D+
66.99	63	D
62.99	60	D-
59.99	0	F

Course Policy:

- **Attendance:** Students are expected to attend all scheduled lectures in this course. Attendance is vital for understanding course material, participating in class discussions, and achieving academic success. Students are responsible for what they miss if they don't attend any of the scheduled lectures
- **Exams:** All the exams in this course will be given on paper, in the classroom and will be proctored by the instructor. A government photo ID (student ID or driver's license) is required when turning an exam. If any of the scheduled exam dates conflict with your religious observances, you must notify your instructor during the first two weeks of the semester. Drop and withdrawal from courses are student responsibilities. Since two best scores out of three test scores are used to determine the final grade, no makeup exams are allowed (i.e., you can miss a Midterm but will then need to take the final).
- **Milestones Assignments:** This class has five mandatory milestones assignments. The following are their policies:
 - Only assignments uploaded to Canvas will be considered for grading.
 - All assignments are due on the designated due date, as specified by the instructor, and it is the student's responsibility to keep track of all assignment due dates and plan accordingly.
 - Students are responsible for what they turn in. If you submit a blank assignment, it will be graded as such. Always double check your submissions to make sure what you meant to upload is actually what you uploaded.
 - Late homework submissions will be subject to penalties. A 10% deduction will be applied for each day that the assignment is overdue, up to a maximum of three days. After three days, the assignment will be considered as not submitted and will be graded as such. This policy applies only to milestones 1, 2, 3 and 4. Milestone 5 must be submitted on time because it is the last milestone of the semester. It is imperative for students to adhere to the specified deadlines, as failing to do so will result in a reduction of their assignment scores. This policy is designed to emphasize the importance of timely and responsible submission of coursework, while also providing a reasonable window for unforeseen circumstances. Students are strongly encouraged to manage their time effectively to avoid any unnecessary deductions and to ensure the best possible academic outcomes.
- **Class Notes:** Students may not capture audio, photos or video from class sessions on their own devices without the explicit permission of the instructor and everyone present, unless part of a DPRC-authorized accommodation.
- **Office Hours Policies:** Office hours are a valuable resource provided by instructors to support students' learning and offer additional assistance outside of regular lectures. They are intended to foster a deeper understanding of course materials, answer specific questions, provide guidance on assignments, and address any concerns related to the course. It is important to emphasize that office hours are not intended to serve as a replacement for missed lectures. Instructors do not typically use office hours to re-lecture or repeat the content covered during a missed class.
- **Instructor Availability:** I will be unable to respond to emails or messages during weekends/holidays. You can still send an email or message asking for help, but I will be unable to provide a response to your request until the next business day
- **Final Grades:** I create a lot of opportunities for students to do well in this course, and all posted grades are final unless there was a verifiable mistake in the grade calculation. Fairness demands, and university rules, require that all students be marked according to the same standards, so that all students receive the marks they earned. I can't just raise your final grade for unjustifiable reasons, so I will not respond to any such requests.

- **Academic Integrity:** Any form of cheating or plagiarism will incur very serious consequences. Students may not post any course materials to any third-party sites (such as Chegg) or post any recordings, screenshots, audio or chat transcripts in any setting outside the class, and that violations of this are subject to student disciplinary action. Any instances of cheating, deceit, fabrication, forgery, plagiarism, unauthorized altering of records or submitting false documents, unauthorized collaboration, unauthorized submission of work previously given credit, or other forms of academic misconduct will be assigned a grade penalty, likely an F or a grade of zero. Failing one or more assignments or examinations for reasons of academic integrity violations may result in a final class grade of F. Students may not withdraw from classes in which they have committed academic misconduct. Consequences for violations of academic integrity may exceed an F on the assignment, examination, or class as determined by the Academic Integrity Review Committee. Members of our academic community have a responsibility to develop an awareness of academic integrity, to cultivate skills to realize honesty in academic and community work, and to sustain actively academic honor as a core value of our community. Students are expected to engage in behaviors that reflect well upon the university. In addition to attending to one's own actions, the Standards for Student Conduct require that students who witness academic dishonesty notify their faculty/instructor, department chair, or the Office of Student Conduct. Supporting academic integrity enhances the reputation of the University and the value attributed to degrees awarded by the University. To learn more about the university policies regarding academic integrity: <https://conduct.sfsu.edu/academic-dishonesty>.
- **Students with disabilities:** Students with disabilities who need reasonable accommodations are encouraged to contact the instructor. The Disability Programs and Resource Center (DPRC) is available to facilitate the reasonable accommodations process. The DPRC is located in the Student Service Building and can be reached by telephone (voice 415-338-2472, video phone 415-335-7210) or by email (dprc@sfsu.edu).
- **SF State fosters a campus free of sexual violence** including sexual harassment, domestic violence, dating violence, stalking, and/or any form of sex or gender discrimination. If you disclose a personal experience as an SF State student, the course instructor is required to notify the Title IX Coordinator by completing the report form available at <http://titleix.sfsu.edu>, emailing vpsaem@sfsu.edu or calling 338-2032. To disclose any such violence confidentially, contact: The SAFE Place - (415) 338-2208; http://www.sfsu.edu/~safe_plc/ Counseling and Psychological Services Center - (415) 338-2208; <http://psyserve.sfsu.edu/> For more information on your rights and available resources: <http://titleix.sfsu.edu>
- **The Religious and Cultural Holidays policy (S09-212)** states that the faculty of San Francisco State University shall accommodate students wishing to observe religious and cultural holidays when such observances require students to be absent from class activities. It is the responsibility of the student to inform the instructor, in writing, about such holidays during the first two weeks of the class each semester. If such holidays occur during the first two weeks of the semester, the student must notify the instructor, in writing, at least three days before the date that they will be absent. It is the responsibility of the instructor to make every reasonable effort to honor the student request without penalty, and of the student to make up the work missed.
- **Academic calendar policy:** The Academic Calendar can be found at <https://webapps.sfsu.edu/public/webcal/acadcalendar>